

Onishi Tatsuki

Data Science and AI Innovation Research Promotion Center

Shiga University

1-1-1, Bamba, Hikone, Shiga, 522-8522 Japan

Telephone: +81-749-27-1023

E-mail: bougtour@gmail.com

ORCID: 0000-0001-7261-9062

27 September 2026

The Editor

Structural Change and Economic Dynamics

Dear Editor,

I am pleased to submit for your consideration the enclosed manuscript, "**Investment Gestation Lags, Intangible Capital, and the Measurement of Capital and Productivity in 39 Economies**", for publication in *Structural Change and Economic Dynamics*.

The paper examines a measurement question that sits squarely within the journal's scope: how the structural transformation of investment — the secular shift from plant and machinery toward software, R&D, and other long-gestation intangible assets — distorts the capital stocks and productivity residuals produced by standard national-accounts conventions. We jointly estimate a time-varying investment-to-output lag $\mu(t)$ and a country-specific intangible-capital share β for 39 OECD and middle-income economies over 1970–2019, using a tempo–quantum decomposition transplanted from formal demography. The corrections narrow the gap between production-side and wealth-side national accounts — surviving a strict temporal holdout on the wealth series (fitted on 1995–2014 and evaluated against 2015–2019, the joint model attains a smaller residual than the production-only alternative in 23 of 39 countries) — revise measured capital by a median –4.3 %, and raise log-level TFP and the implied labour share by a median 1.7 percentage points each, with the largest effects precisely where the investment mix has shifted most.

I believe this work is well-suited to *Structural Change and Economic Dynamics* for three reasons.

First, the paper's subject is structural change itself: it quantifies how a documented shift in the composition of capital formation propagates through the measurement system on which the analysis of structural and technological change

depends. Readers of the journal working on productivity dynamics, intangible-driven growth, or the changing configuration of national accounts will find directly usable estimates.

Second, the analysis contributes econometric tools designed for the study of structural dynamics. A time-varying gestation lag $\mu(t)$ is identified jointly with the intangible share β by a flow–stock consistency constraint; the "Relational PIM" adapts Brass relational models to capital accounting; and an observable, parameter-free tempo proxy M_{obs} is constructed from OECD asset-composition data, so the core correction can be replicated at negligible cost by any statistical agency.

Third, the reporting is deliberately conservative: the out-of-sample improvement of the tempo correction (median MAPE 4.60 % $\rightarrow$ 3.99 %) is reported together with its non-significant Wilcoxon test, and robustness is documented extensively in the Supporting Information — Monte Carlo identification checks, depreciation-sensitivity, R&D-scaling sensitivity, and historical-episode decompositions.

I confirm the following:

- This work is original and has not been published elsewhere.
- The manuscript is not under consideration by any other journal.
- All data sources (Penn World Table 10.01; World Bank Changing Wealth of Nations; World Bank WDI; OECD GFCF) are publicly available; frozen extracts, code, and intermediate outputs are archived in the companion repository at <https://github.com/bougtoir/gdp-tempo-paper>, and a one-command reproduction pipeline (`make`) regenerates every figure, table, and number in the manuscript.
- I am the sole author.
- I have no competing interests, financial or otherwise, related to this work.
- I used generative AI to assist with formatting the text, choosing words, and writing analysis code; this is documented in the Declarations section of the title page.
- I am aware of the journal's submission fee (USD 100) and will complete payment in Editorial Manager upon submission.

A title page with full author details and a Declarations section is included as a separate file.

I would be grateful for your consideration.

Yours sincerely,

Onishi Tatsuki

Data Science and AI Innovation Research Promotion Center

Shiga University

bougtoir@gmail.com